

Scala For DS

Practical no:-21

Roll No. : S019	Name: SUMEET KUMAVAT
Class : SY BSC IT	Date/Time of Submission : 25-08-2026

Aim:-Join two CSV files in Spark DataFrames based on a common column and write the output to a file.

Code:-

```
build.sbt  file1.csv  file2.csv  Practical21.scala ×
1  import org.apache.spark.sql.SparkSession
2
3  object Practical21 extends App { new *
4      val s = SparkSession.builder()
5          .appName("JoinTwoCSV")
6          .master("local[*]")
7          .getOrCreate()
8
9      val d1 = s.read
10         .option("header", "true")
11         .csv("File1.csv")
12
13     val d2 = s.read
14         .option("header", "true")
15         .csv("File2.csv")
16
17     val j = d1.join(d2, Seq("id"), "inner")
18     j.show()
19     j.write.option("header", "true")
20     s.stop()
21 }
22
```

Output:-

```
+---+-----+-----+
| id|  name|score|
+---+-----+-----+
| 1| Sumeet| 90|
| 2|Harshal| 70|
| 3| Dhruv| 85|
| 4| Niraj| 80|
+---+-----+-----+
```

```
Sumeet S019
```

```
Process finished with exit code 0
```

Scala For DS

Practical no:-22

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Aim:- Set up Apache Spark locally and count the frequency of words in a text file.

CODE:-

```
build.sbt  Practical22.scala  input.txt
1  import org.apache.spark.sql.SparkSession
2
3  object Practical22 extends App {
4      val s = SparkSession.builder()
5          .appName("WordCountExample")
6          .master("local[*]")
7          .getOrCreate()
8
9      val r = s.sparkContext.textFile("input.txt")
10     val w = r.flatMap(line => line.split("\\s+"))
11     val wo = w.map(word => (word, 1))
12     val wc = wo.reduceByKey(_ + _)
13     wc.collect().foreach(println)
14     s.stop()
15 }
16
```

OUTPUT:-

```
25/08/25 10:54:01 AM
(is,2)
(easy,1)
(powerful,1)
(Spark,2)
25/08/25 10:54:01 AM
```

Scala For DS

Practical no:-23

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Aim:- Filter rows in a CSV file using Spark DataFrames where a numeric column exceeds a threshold.

CODE:-

```
build.sbt  Practical22.scala  Practical23.scala ×  superstore.csv
1      import org.apache.spark.sql.SparkSession
2
3  ▶    object Practical23 extends App { new *
4
5        val spark = SparkSession.builder()
6          .appName("FilterCSVExample")
7          .master("local[*]")
8          .getOrCreate()
9
10     import spark.implicits._
11     val df = spark.read
12       .option("header", "true")
13       .option("inferSchema", "true")
14       .csv("src/main/resources/superstore.csv")
15
16     val f = df.filter($"Sales" > 5000)
17     println("Filtered Data:")
18     f.show()
19     spark.stop()
20   }
```

OUTPUT:-

20/09/20 11:27:22 INFO SparkContext: SparkContext is stopping with exitcode 0.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central
510	CA-2015-145352	3/16/2015	3/22/2015	Standard Class	CM-12385	Christopher Martinez	Consumer	United States	Atlanta	Georgia	30318	South
684	US-2017-168116	11/4/2017	11/4/2017	Same Day	GT-14635	Grant Thornton	Corporate	United States	Burlington	North Carolina	27217	South
2506	CA-2014-143917	7/25/2014	7/27/2014	Second Class	KL-16645	Ken Lonsdale	Consumer	United States	San Francisco	California	94122	West
2624	CA-2017-127180	10/22/2017	10/24/2017	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	New York	10024	East
2698	CA-2014-145317	3/18/2014	3/23/2014	Standard Class	SM-20320	Sean Miller	Home Office	United States	Jacksonville	Florida	32216	South
4099	CA-2014-116904	9/23/2014	9/28/2014	Standard Class	SC-20095	Sanjit Chand	Consumer	United States	Minneapolis	Minnesota	55407	Central
4191	CA-2017-166709	11/17/2017	11/22/2017	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Newark	Delaware	19711	East
4278	US-2016-107440	4/16/2016	4/20/2016	Standard Class	BS-11365	Bill Shonely	Corporate	United States	Lakewood	New Jersey	8701	East
6341	CA-2017-143112	10/5/2017	10/9/2017	Standard Class	TS-21370	Todd Sumrall	Corporate	United States	New York City	New York	10035	East
6426	CA-2016-143714	5/23/2016	5/27/2016	Standard Class	CC-12370	Christopher Conant	Consumer	United States	Philadelphia	Pennsylvania	19120	East
6521	CA-2017-138289	1/16/2017	1/18/2017	Second Class	AR-10540	Andy Reiter	Consumer	United States	Jackson	Michigan	49201	Central
6627	CA-2014-145541	12/14/2014	12/21/2014	Standard Class	TB-21400	Tom Boeckenhauer	Consumer	United States	New York City	New York	10024	East
6827	CA-2016-118689	10/2/2016	10/9/2016	Standard Class	TC-20980	Tamara Chand	Corporate	United States	Lafayette	Indiana	47905	Central
7667	US-2016-140158	10/4/2016	10/8/2016	Standard Class	DR-12940	Daniel Raglin	Home Office	United States	Providence	Rhode Island	2908	East
8154	CA-2017-140151	3/23/2017	3/25/2017	First Class	RB-19360	Raymond Buch	Consumer	United States	Seattle	Washington	98115	West
8489	CA-2016-158841	2/2/2016	2/4/2016	Second Class	SE-20110	Sanjit Engle	Consumer	United States	Arlington	Virginia	22204	South
8859	CA-2017-135909	10/13/2017	10/20/2017	Standard Class	JW-15220	Jane Waco	Corporate	United States	Sacramento	California	95823	West
9040	CA-2016-117121	12/17/2016	12/21/2016	Standard Class	AB-10105	Adrian Barton	Consumer	United States	Detroit	Michigan	48205	Central

Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Central	TEC-MA-10000822	Technology	Machines	Lexmark MX611dhe ...	8159.952	8	0.4	-1359.992
South	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 EL...	6354.95	5	0	3177.475
South	TEC-MA-10004125	Technology	Machines	Cubify CubeX 3D P...	7999.98	4	0.5	-3839.9904
West	OFF-SU-10000151	Office Supplies	Supplies	High Speed Automa...	8187.65	5	0	327.506
East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	11199.968	4	0.2	3919.9888
South	TEC-MA-10002412	Technology	Machines	Cisco TelePresenc...	22638.48	6	0.5	-1811.0784
Central	OFF-BI-10001120	Office Supplies	Binders	Ibico EPK-21 Elec...	9449.95	5	0	4630.4755
East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	10499.97	3	0	5039.9856
East	TEC-MA-10001047	Technology	Machines	3D Systems Cube P...	9099.93	7	0	2365.9818
East	TEC-MA-10001047	Technology	Machines	3D Systems Cube P...	5199.96	4	0	1351.9896
East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	8399.976	4	0.4	1119.9968
Central	OFF-BI-10004995	Office Supplies	Binders	GBC DocuBind P400...	5443.96	4	0	2504.2216
East	TEC-MA-10001127	Technology	Machines	HP Designjet T52...	6999.96	4	0	2239.9872
Central	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	17499.95	5	0	8399.976
East	TEC-CO-10001449	Technology	Copiers	Hewlett Packard L...	5399.91	9	0	2591.9568
West	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	13999.96	4	0	6719.9808
South	TEC-MA-10001127	Technology	Machines	HP Designjet T52...	8749.95	5	0	2799.984
West	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 EL...	5083.96	5	0.2	1906.485
Central	OFF-BI-10000545	Office Supplies	Binders	GBC Ibimaster 500...	9892.74	13	0	4946.37

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Practical no:-24

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Aim:- Perform a group-by operation in Spark DataFrames to compute the average of each group.

CODE:-

```
build.sbt  Practical22.scala  Practical23.scala ×  superstore.csv  input.txt
1  import org.apache.spark.sql.SparkSession
2  import org.apache.spark.sql.functions._
3
4  ▶ object Practical23 extends App { new *
5
6      val spark = SparkSession.builder()
7          .appName("AverageSales")
8          .master("local[*]")
9          .getOrCreate()
10
11     val df = spark.read
12         .option("header", "true")
13         .option("inferSchema", "true")
14         .csv("src/main/resources/superstore.csv")
15
16     val result = df.groupBy("Category")
17         .agg(avg("Sales").alias("Average Sales"))
18
19     println("Average Sales by Category:")
20     result.show()
21     spark.stop()
22 }
```

OUTPUT:-

```
20/08/20 11:31:00 INFO SparkUI: Stopped Spark w
+-----+-----+
|      Category|      Average Sales|
+-----+-----+
|Office Supplies|121.69225531914947|
|      Furniture| 353.4459311957568|
|      Technology| 454.5405475802047|
+-----+-----+
```